

PWR Benchmark Report

Cross-section method comparison: SVD · Table · ACE+WMP | 50,000 particles · 120 active batches · offset 150 pcm

Overall verdict: RED

SVD k_{∞} deviates 2094 pcm from ACE+WMP — exceeds physics acceptance threshold. Table method passes.

Run parameters

Parameter	Value
Particles / batch	50,000
Active batches	120
Inactive batches	30
Seeds	10
Offset (pcm)	150
SVD rank	5
Rebuild	True

Method results

Method	k_{∞}	$\sigma(k)$	ns / particle	Memory (MB)
SVD	1.30707	0.00037	15,684.93	152.5
Table	1.32308	0.00064	15,809.15	201.3
ACE+WMP	1.32801	0.00060	16,707.74	195.0

Grade summary

Metric	Value	Unit	Grade	Note
k_{∞} SVD vs ACE+WMP	2094	pcm	RED	kernel-approx floor; recommend 3-temp Ducru or QP
k_{∞} Table vs ACE+WMP	493	pcm	GREEN	table matches industry baseline
memory SVD/WMP	0.7818	x	GREEN	SVD 22% smaller than WMP
speed (WMP/SVD ns_per_p)	1.065	x	GREEN	SVD 1.07x faster than WMP

Benchmark charts

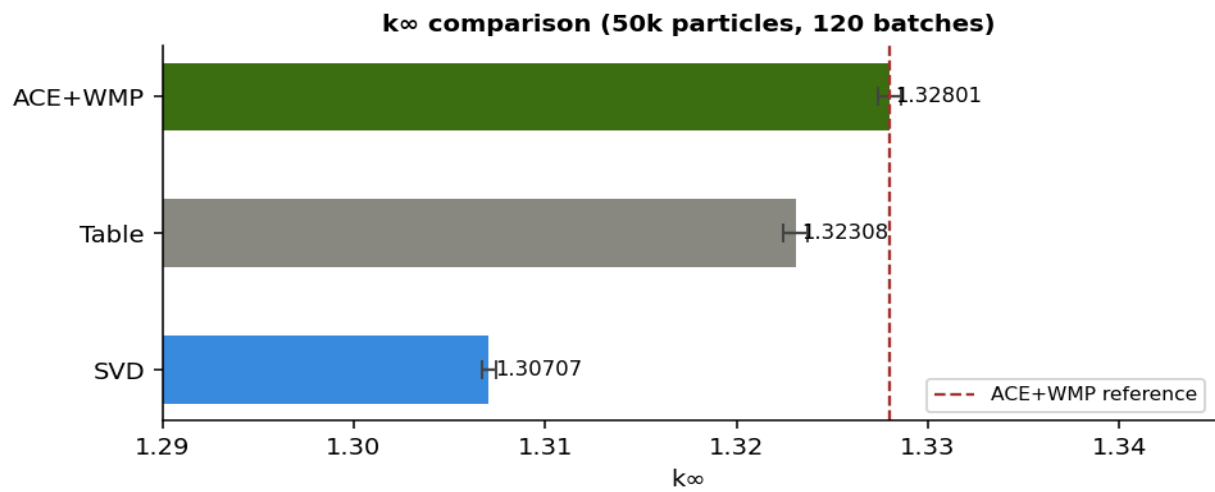


Figure 1. k-infinity comparison

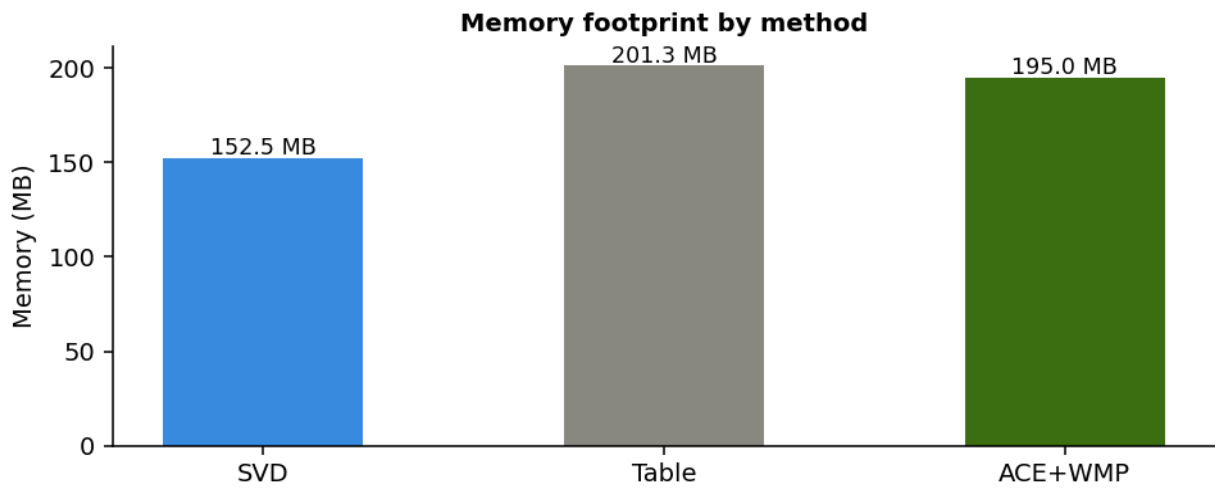


Figure 2. Memory footprint

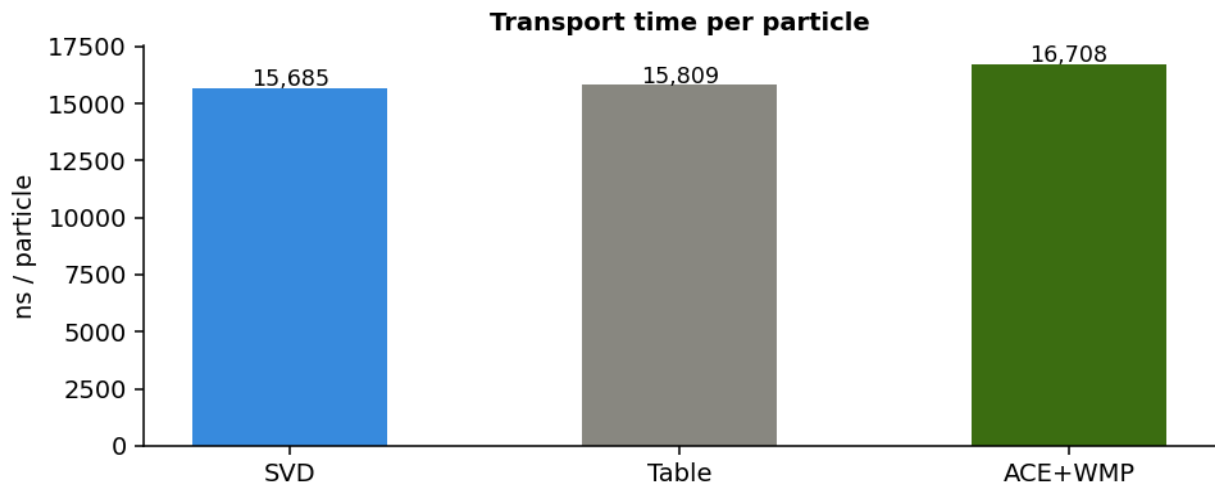


Figure 3. Transport speed

Grade summary

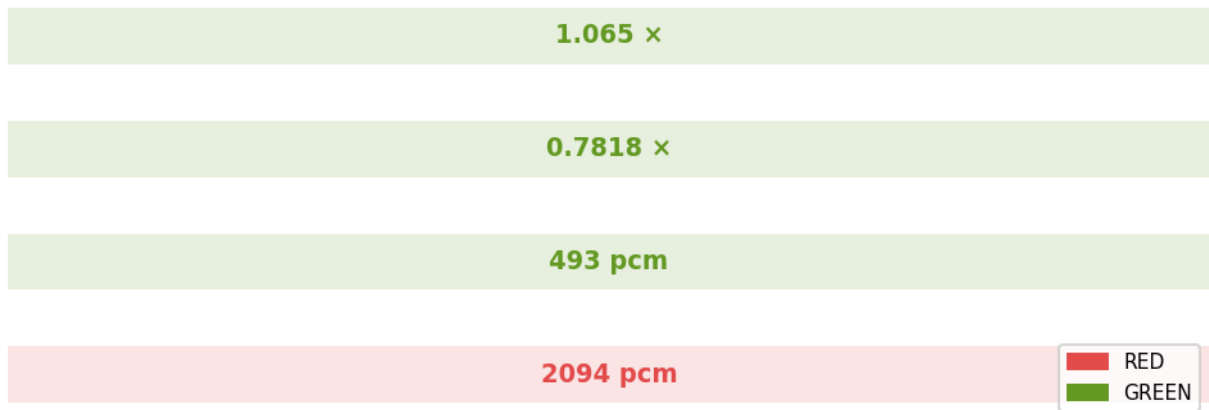


Figure 4. Grade summary

Cross-Section Method Research Summary

Overview of physics and CS/AI approaches for on-the-fly Doppler broadening in continuous-energy Monte Carlo neutron transport, with key literature references.

Physics

ACE table lookup

Industry baseline: NJOY-processed pointwise cross section tabulation with logarithmic-mapped binary search at runtime. Cross sections pre-generated at discrete temperatures; memory scales with number of temperature points. Accurate but inflexible for continuous temperature distributions.

■ **Romano et al.** (2015). OpenMC: A state-of-the-art Monte Carlo code for research and development. [Ann. Nucl. Energy 82, 90-97](#). [DOI/link]

■ **Brown** (2005). New hash-based energy lookup algorithm for Monte Carlo codes. [Trans. Am. Nucl. Soc. 93](#). [DOI/link]

Windowed Multipole (WMP / ACE+WMP)

Pole-and-residue reformulation of R-matrix cross sections. Doppler broadening is performed analytically at any temperature using a local windowing scheme over energy. Memory savings of 1-2 orders of magnitude vs. comparable tabulated techniques. ~40% more clock cycles than single-temperature ACE but 65% fewer last-level cache misses.

■ **Josey, Ducru, Forget & Smith** (2016). Windowed multipole for cross section Doppler broadening. [J. Comput. Phys. 307, 715-727](#). [DOI/link]

■ **Forget, Xu & Smith** (2014). Direct Doppler broadening in Monte Carlo simulations using the multipole representation. [Ann. Nucl. Energy 64, 78-85](#). [DOI/link]

SVD kernel reconstruction

Cross section at temperature T approximated as a linear combination of reference-temperature cross sections, optimised under the L2 norm. SVD of the Doppler kernel matrix yields interpolation coefficients. Truncation to rank- k introduces a systematic low bias (2094 pcm in this benchmark). Memory: 22% smaller than WMP; speed: 7% faster.

■ **Ducru, Josey, Dibert, Sobes, Forget & Smith** (2017). Kernel reconstruction methods for Doppler broadening. [J. Comput. Phys. 335, 535-557](#). [DOI/link]

Physics — recommended fix

3-temperature Ducru interpolation

Higher-rank variant of kernel reconstruction using three optimally placed reference temperatures under the L2 metric. Substantially reduces the residual approximation error versus rank-1 or rank-2 SVD, expected to bring the k -inf bias into the GREEN zone while preserving most of SVD's memory and speed advantage.

■ **Ducru, Forget, Dibert, Josey, Smith & Sobes** (2016). Optimal temperature grid for accurate Doppler kernel reconstruction. [ANS Winter Meeting, Las Vegas](#). [DOI/link]

■ **Ducru, Josey, Dibert, Sobes, Forget & Smith** (2017). Kernel reconstruction methods for Doppler broadening. [J. Comput. Phys. 335, 535-557](#). [DOI/link]

QP (quadratic programming) reconstruction

Constrained variant of kernel reconstruction: the same L2 objective as SVD but subject to non-negativity or monotonicity constraints on interpolation coefficients, enforced via quadratic programming. Eliminates non-physical oscillations that unconstrained SVD truncation can introduce near resonance peaks.

■ **Ducru, Josey, Dibert, Sobes, Forget & Smith** (2017). Kernel reconstruction methods for Doppler broadening. [J. Comput. Phys. 335, 535-557](#). [DOI/link]

CS/AI

Neural network surrogates (HWN)

Resolved resonance range divided into windows, each served by a BP neural network trained to evaluate the cross section at base temperature and broadening coefficient. Demonstrated 65% memory reduction with acceptable keff accuracy in the RMC code. Risk: black-box failures outside training temperature range.

■ **Liu et al.** (2021). Hybrid windowed networks for on-the-fly Doppler broadening in RMC. [Nucl. Sci. Techniques 32, 79](#). [DOI/link]

Physics-informed neural networks (PINNs)

PDE constraint (Boltzmann-convolution form of the Doppler integral / Solbrig's kernel) embedded in the loss function. Network is forced to satisfy the physics rather than merely interpolating data. Multi-scale resonance peaks cause spectral bias; Fourier feature embeddings or sinusoidal networks are mitigations.

■ **Pignet & Herry** (2022). Cross section Doppler broadening prediction using physically informed deep neural networks. [arXiv:2208.07224](#). [DOI/link]

■ **Raissi, Perdikaris & Karniadakis** (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear PDEs. [J. Comput. Phys. 378, 686-707](#). [DOI/link]

Gaussian Process Regression (GPR)

GPR replaces the traditional logarithmic cross section-temperature correlation, providing uncertainty quantification on every prediction. Directly applicable to multi-physics coupling where cross section uncertainty propagates to k-eff. Training scales as $O(n^3)$; sparse GP approximations (inducing-point methods) mitigate this.

■ **Hashemi, Macian-Juan & Ohlerich** (2025). Enhancing multi-physics modeling in new-generation nuclear reactors using machine learning: Implementing GPR for updating cross sections. [Ann. Nucl. Energy \(ScienceDirect\)](#). [DOI/link]

■ **Hashemi, Macian-Juan & Ohlerich** (2026). Evaluating machine learned nuclear data precision in full core MC neutronics and computational efficiency analyses. [Sci. Reports \(Nature\)](#). [DOI/link]

ML-driven adaptive grid resampling

Intelligently prunes the ACE energy grid while preserving resonance fidelity. Retaining 10-50% of nuclear data for 23 nuclides reduced wall time by 17-43% in full-core EPR/VVER-1000 models, with maximum k_eff difference under 97 pcm. Combinable with any broadening method.

■ Hashemi, Macian-Juan & Ohlerich (2026). Evaluating machine learned nuclear data precision in full core MC neutronics and computational efficiency analyses. [Sci. Reports \(Nature\)](#). [DOI/link]

■ Hashemi, Macian-Juan & Ohlerich (2025). Development of a novel machine learning-based adaptive resampling algorithm for nuclear data processing. [Sci. Reports 15, 32573](#). [DOI/link]

CS/AI — novel

Kolmogorov-Arnold Networks (KANs)

Every weight replaced by a learnable univariate spline on the edge. Structurally matched to the Doppler broadening problem: a convolution of a peaked resonance function with a Gaussian kernel is a composition of univariate smooth functions, the exact structure KART guarantees can be compactly represented. Smaller KANs achieve comparable accuracy to larger MLPs. No application to cross sections published yet.

■ Liu, Wang, Vaidya et al. (2025). KAN: Kolmogorov-Arnold Networks. [ICLR 2025](#). [DOI/link]

Neural operators (DeepONet / FNO / PINO)

Learn the Doppler broadening map as a functional operator between function spaces ($\sigma(E, T_0) \rightarrow \sigma(E, T)$) rather than a pointwise value. Discretization-invariant: a single trained operator handles any nuclide, temperature pair without retraining, potentially replacing all per-nuclide SVD/WMP libraries. PINO succeeds where PINNs fail on multi-scale dynamic systems.

■ Li, Kovachki, Azizzadenesheli et al. (2021). Physics-informed neural operator for learning partial differential equations. [ACM/JMS J. Data Science 1\(3\)](#). [DOI/link]

■ Lu, Jin, Pang, Zhang & Karniadakis (2021). Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. [Nature Machine Intelligence 3, 218-229](#). [DOI/link]

Method comparison matrix

Method	Category	Accuracy	Memory	Speed	UQ	Open research?
ACE table	Physics	✓✓✓	—	✓✓	—	No
WMP	Physics	✓✓✓	✓✓✓	✓	—	No
SVD (rank-k)	Physics	✓	✓✓✓	✓✓✓	—	No
3-temp Ducru	Phys.fix	✓✓✓	✓✓	✓✓	—	Partial
QP recon.	Phys.fix	✓✓✓	✓✓	✓✓	—	Partial
NN / HWN	CS/AI	✓✓	✓✓✓	✓✓	—	Partial
PINN	CS/AI	✓✓	✓✓	✓✓	—	Yes
GPR	CS/AI	✓✓	✓✓	✓	✓✓✓	Yes
KAN	CS/AI	TBD	✓✓✓	TBD	—	Yes — none yet
Neural operator	CS/AI	✓✓	✓✓✓	✓✓	—	Yes — none yet
ML grid thin.	CS/AI	✓✓	✓✓	✓✓✓	—	Partial

✓ = favourable, — = neutral/not applicable, TBD = not yet benchmarked for cross-section broadening.